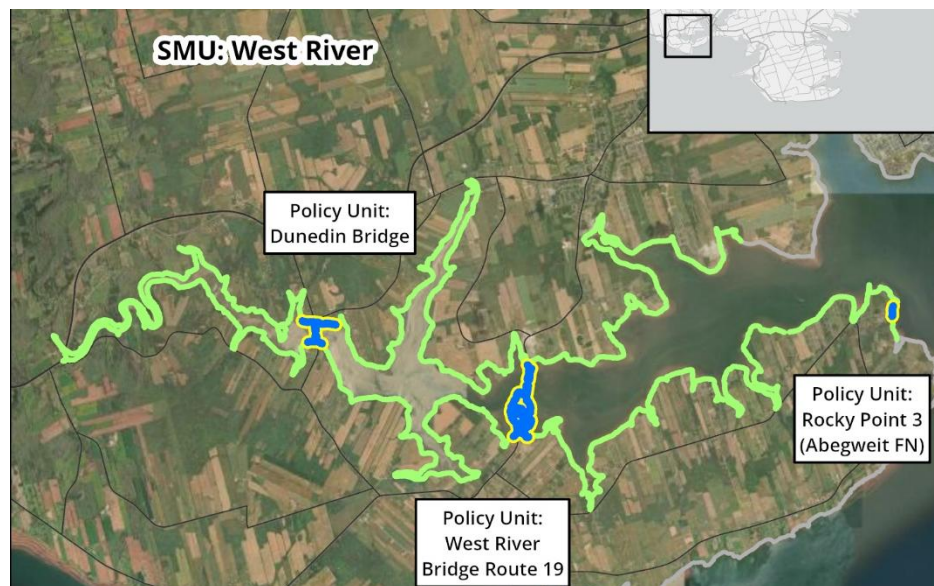


10. West River

Shoreline length: 104 km



Shore Types

West River is an estuary located on the southeast end of PEI. Shore type is characterized by a majority of wetland and low plains.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
41	9	6	2	42	0	0	0



Rocky Point shoreline



Drakes Point



Route 19 bridges

Governance, Infrastructure, and Economy

Approximately 87% of this area falls within Cornwall and West River; the remaining area includes Clyde River (~9%) and Rocky Point Indian Reserve No. 3, under the jurisdiction of the Abegweit First Nation (~4%).

Key infrastructure includes various bridges along with Rock Point, Route 19 at Canoe Cove, Black, Cody's, Wilson's, and MacEwan's Creeks and bridges; West River Road (Route 9) at Long Creek; the causeway/bridge to Dunedin; and the Cornwall wastewater treatment lagoons (Hyde Pond).

The area has three major golf courses as key tourism and recreation drivers of the economy: the Clyde River Golf Course, Countryview Golf Course, along with the Feehan's Creek Mobile Home Park. Local and legacy land use constraints include some areas with small lots, a few unpaved roads, and some undeveloped small-lot subdivisions with undeveloped or unpaved roads. There are many small residential lots at Rocky Point, but otherwise the area sees little development; no coastal roads or facilities are otherwise identified, and there are no major coastal economic drivers.

Natural Assets and Heritage

The area is home to Ferry Road Centre Park, Strathgartney and Bonshaw Hills Provincial Park, and Hyde Creek Natural Area, which include playgrounds, dog parks, and active transportation trails.

The shoreline includes critical habitat for bird species including the threatened Bank Swallow.

Planning and Policy Framework

Cornwall has municipal planning authority, applying an Environmental Reserve Overlay zone (O2) that includes watercourses and wetlands plus a 15m buffer; the minimum setback is the greater of 15m or 60× the annual erosion rate, where applicable. An additional lot setback is also required. A vertical setback is included, based on the 2100 coastal floodplain. West River has municipal planning authority, applying a hard Environmental Risk zone that includes watercourses and wetlands plus 15m and the 2100 coastal floodplain, as well as other hazard areas. All building setbacks are measured from the zone boundary. Coastal setbacks are based on standard buffers plus setbacks, or 60× the annual rate of erosion where applicable. A vertical setback is included, based on the 2100 coastal floodplain. Clyde River is a municipality with no planning; provincial rules apply, including horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted. The Abegweit First Nation manages land use within Rocky Point Indian Reserve No. 3 under its own authority, separate from municipal or provincial planning.

Sediment transport

Sediment transport rates and associated downdrift risk across the SMU are low.

Coastal Flood and Erosion Risk

Coastal flooding is the prevailing hazard for the area. The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline].

	FLOOD RISK				EROSION RISK		
	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)	Short-term (~2050)	Medium-term (~2080)	Long-term (~2130)
Buildings	33 [<1/km]	44 [<1/km]	57 [1/km]	87 [1/km]	4 [<1/km]	6 [<1/km]	18 [<1/km]
Roads	1073 m [10 m/km]	1780 m [17 m/km]	2910 m [28 m/km]	3992 m [39 m/km]	196 m [2 m/km]	287 m [3 m/km]	677 m [7 m/km]

Proposed SMP Strategies

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
10.00 West River	NAI	MR	MR
Rationale	Given the initially low risk levels the proposed strategy is No Active Intervention. Mid- to long-term Managed Realignment will reduce flood risk to the limited number of vulnerable assets while preserving natural shorelines, including wetlands and low plains.		
10.01 Dunedin Bridge -PU	R	R	R
Rationale	Resist strategy would be suitable for the shorelines around the bridge which is a key infrastructure asset to the community.		
10.02 Rocky Point 3 (Abegweit FN) -PU	R	R	R
Rationale	The Resist strategy is suggested based on initial review of shoreline and risk information. This would include maintenance of existing revetments. Flood hazard is limited to area surrounding the pond north of Harbour View Dr. Final strategies and local policies will be developed with the local authority.		
10.03 West River bridge Route 19 -PU	R	R	R
Rationale	Resist strategy would be suitable for the bridges and adjacent shores, but will need to incorporate flood mitigation.		

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.